

FUNCTIONS

Dr. Joshua Kiddy Kwasi Asamoah

Department of Mathematics
Kwame Nkrumah University of Science and Technology



Lecture Outline

- 1 Introduction
- 2 Constant, Step, and Piecewise functions
 - Constant Function
 - Step Function
 - Piecewise Functions
- 3 Power functions
 - Monomials: $p \in \mathbb{Z}_+$
 - Reciprocal functions: $p \in \mathbb{Z}_-$
 - Radical functions: p form is $1/n$; $n \in \mathbb{N}$
- 4 Polynomial Functions
- 5 Rational Functions
- 6 Algebraic Functions
- 7 Transcendental Functions
 - Exponential Functions
 - Logarithmic Functions
 - Trigonometric Functions



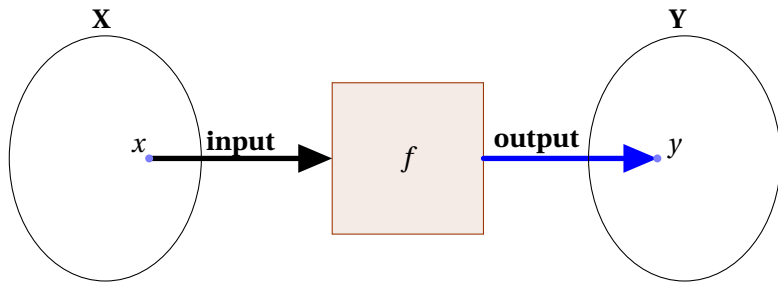
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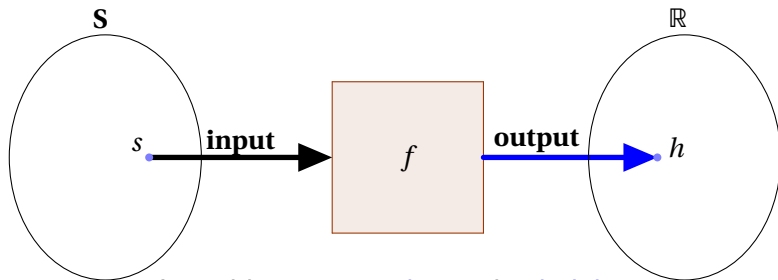
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- 2 In general, a function is a map between two sets that assigns to each element in the first set a **unique** element in the second set.



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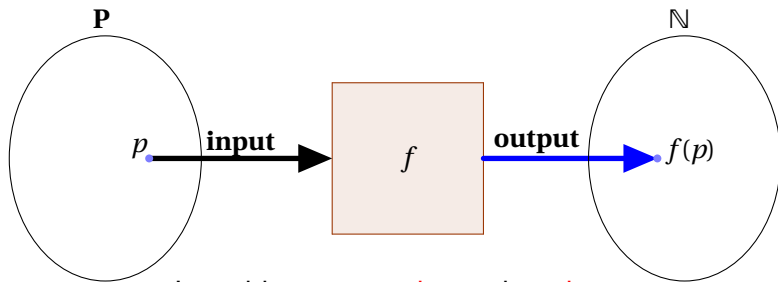


It could map a **student** to its **height**.



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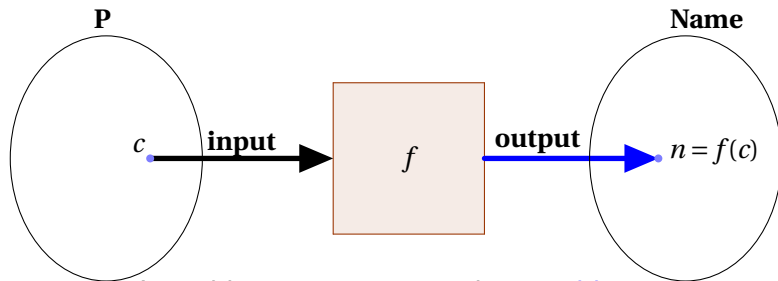


It could map a **product** to its **price**.



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It could map a **country** to its **president**.



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Remark

- ① Uniqueness here means **an input** cannot yield more than one output i.e. $x \mapsto y_1, y_2$ is not allowed.
- ② However, **two different inputs** x_1 and x_2 can be assigned to the same output y .

Definition: If $x \rightarrow f \rightarrow y$

Definition (Domain)

The **domain** D_f of a function f , is the set of all possible inputs where f is defined.



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The set of all assigned outputs, $f(X) = \{ f(x) \mid x \in X \}$, is called the **range/image** of f .



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Definition (Graph)

The graph of the function $y = f(x)$ is a pictorial representation of the function. It is the collection of the points $(x, f(x))$. It represents a curve in the Cartesian plane.

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Constant functions

$$\begin{array}{rcl} f : & \mathbb{R} & \rightarrow \mathbb{R} \\ & x & \mapsto c \end{array}$$

- ① Any real number x is assigned to the **unique real number c** : f is a function, but specifically, f is a **constant function**.



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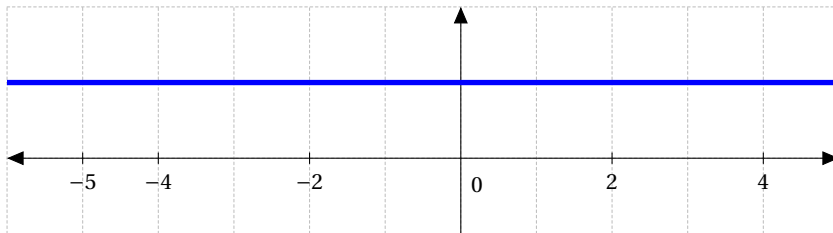


Constant function graph

Definition

A constant function is a function whose value is the same for every input value

$$\begin{aligned} f: \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto 1 \end{aligned}$$



Step Function (or staircase function)

Definition

They are function that increases or decreases abruptly from one constant value to another.

$$\begin{aligned} f: \mathbb{R} &\rightarrow (-\infty, 10) \\ x &\mapsto f(x) = \begin{cases} 2 & \text{if } x \leq -4 \\ 1 & \text{if } -4 < x \leq 0 \\ -1 & \text{if } 0 < x \leq 2. \end{cases} \end{aligned}$$



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- 3 Its range is $\{-1, 1, 2\}$ and its codomain is $(-\infty, 10)$.



Step Function graph

$$f: \mathbb{R} \rightarrow (-\infty, 10)$$

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Example of Non Step Function

Consider the following relation

$$g: \mathbb{R} \rightarrow \mathbb{R}$$
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- 1 g assigns two values to $x_0 = -4$.
- 2 That is $g(-4) = \{1, 2\}$.
- 3 Thus, g is **NOT** a function.



Piecewise functions

Definition

Piecewise functions are defined by different functions for different intervals of the domain.

$$\begin{aligned} f: \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto f(x) = \begin{cases} -x+2 & \text{if } x \geq 0 \\ 2x+2 & \text{if } x < 0. \end{cases} \end{aligned}$$

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- 3 Its range is $I = \{ -x+2 \mid x \geq 0 \} \cup \{ 2x+2 \mid x < 0 \}$.

$$x \geq 0 \implies -x \leq 0 \implies -x+2 \leq 2.$$

$$x < 0 \implies 2x+2 < 2.$$

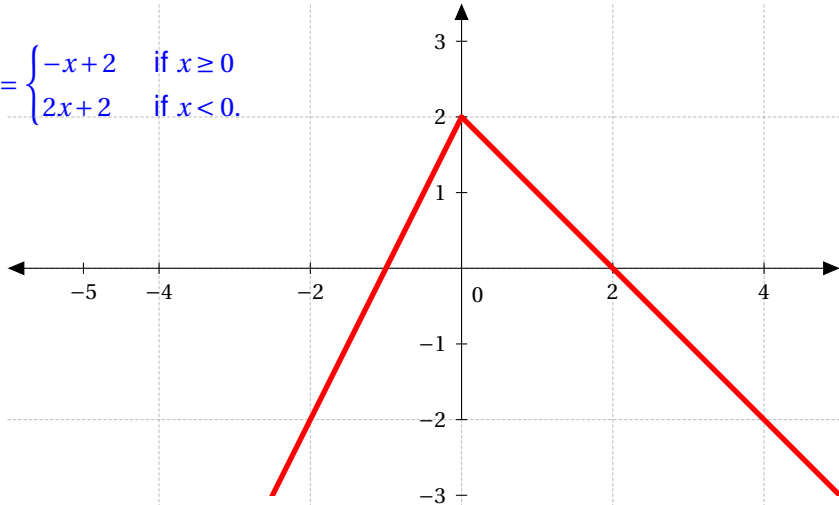
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Piecewise functions graph

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Power Functions

Definition

Power functions are functions of the form of

$$f(x) = ax^p \tag{1}$$

where p is any real number ($p \in \mathbb{R}$) and a is a non-zero real number, that is ($a \in \mathbb{R} - \{0\}$).



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Some examples of Power Functions

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- ② **Reciprocal Functions** are power function with negative power, that is $p \in \mathbb{Z}_-$
- ③ **Radical Functions** are power functions where the degree p is of the form $\frac{1}{n}$ and $n \in \mathbb{N}$



Power Functions

Example

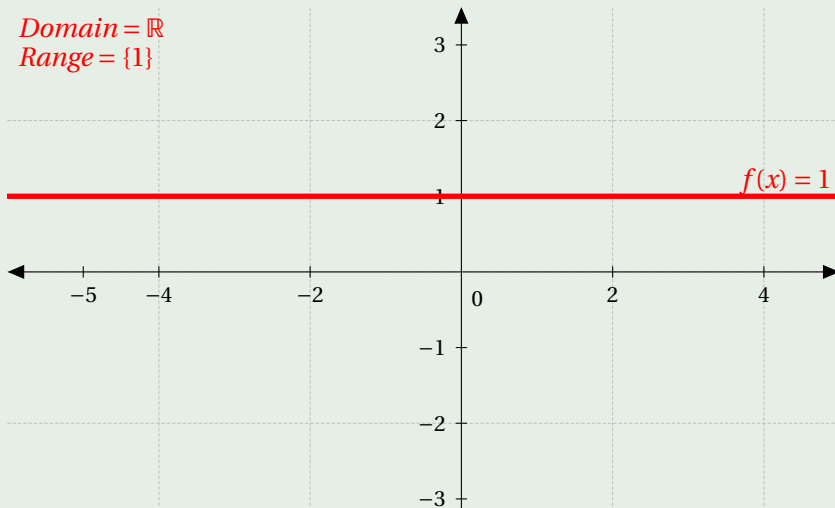
$f(x)$	Domain	Range
x^{2n}	$\mathbb{R}$	$\mathbb{R}_+$
x^{2n+1}	$\mathbb{R}$	$\mathbb{R}$
$\frac{1}{x^{2n}}$	$\mathbb{R} - \{0\}$	$\mathbb{R}_+ - \{0\}$
$\frac{1}{x^{2n+1}}$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$

$f(x)$	Domain	Range
$\sqrt[2n]{x}$	$\mathbb{R}_+$	$\mathbb{R}_+$
$\sqrt[2n+1]{x}$	$\mathbb{R}$	$\mathbb{R}$
$\frac{1}{\sqrt[2n]{x}}$	$\mathbb{R}_+ - \{0\}$	$\mathbb{R}_+ - \{0\}$
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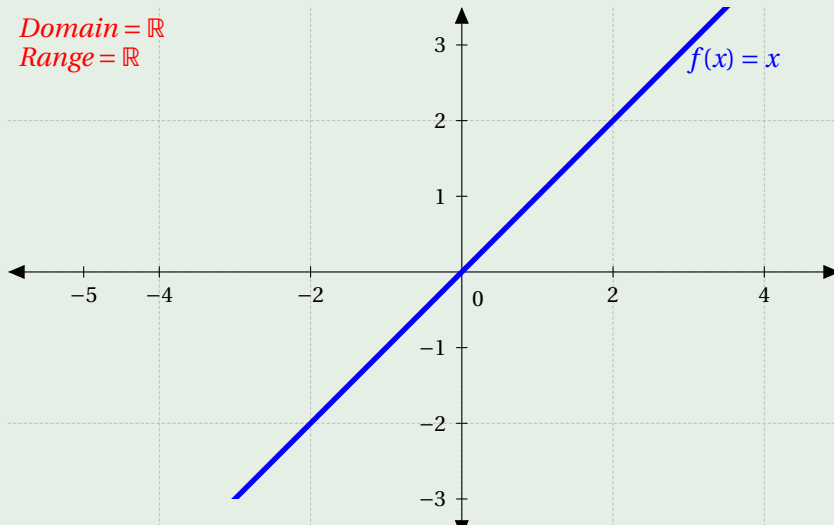
$n \in \mathbb{Z}_+$, $\mathbb{R} - \{0\} = (-\infty, 0) \cup (0, +\infty)$, $\mathbb{R}_+ = [0, +\infty)$, $\mathbb{R}_+ - \{0\} = (0, +\infty)$

Monomials: Graph of $f(x) = 1$

Domain = $\mathbb{R}$
Range = $\{1\}$

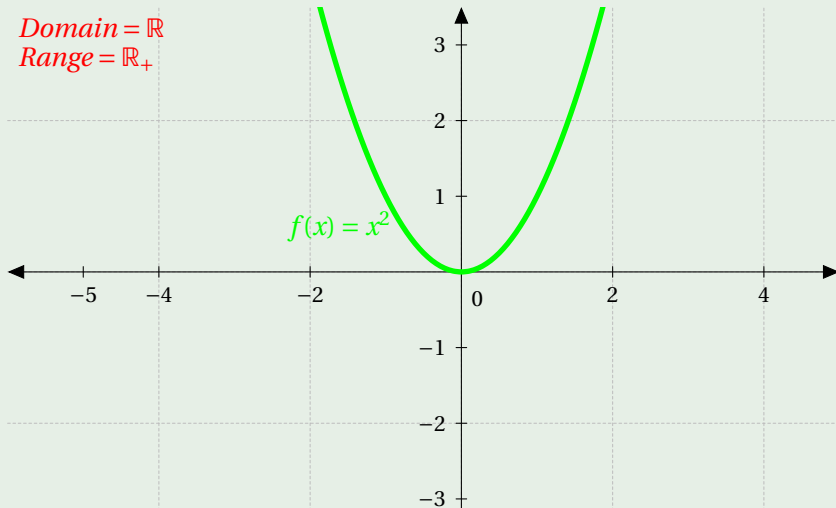


Monomials: Graph of $f(x) = x$



Monomials: Graph of $f(x) = x^2$

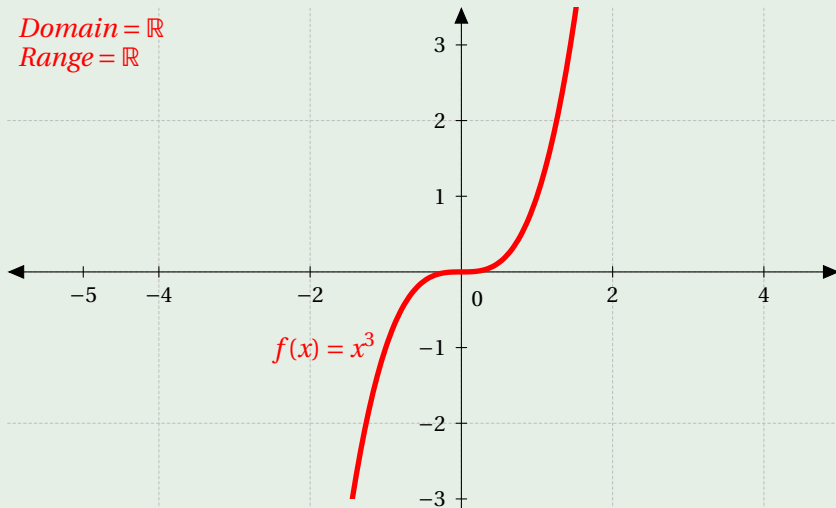
Domain = $\mathbb{R}$
Range = $\mathbb{R}_+$



Monomials: Graph of $f(x) = x^3$

Domain = $\mathbb{R}$

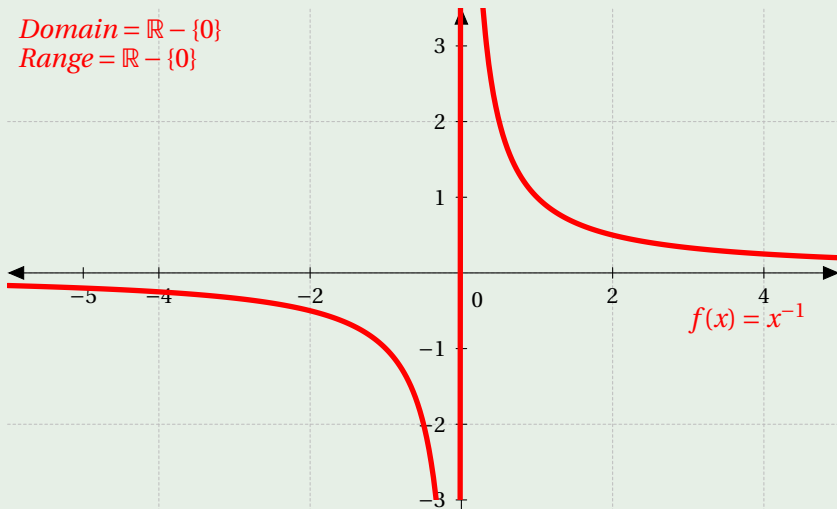
Range = $\mathbb{R}$



Reciprocal functions: Graph of $f(x) = x^{-1}$

Domain = $\mathbb{R} - \{0\}$

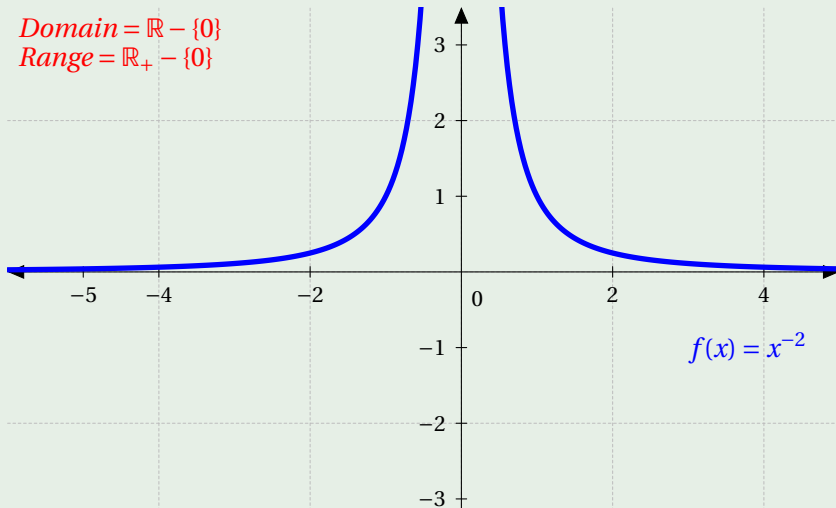
Range = $\mathbb{R} - \{0\}$



Reciprocal functions: Graph of $f(x) = x^{-2}$

Domain = $\mathbb{R} - \{0\}$

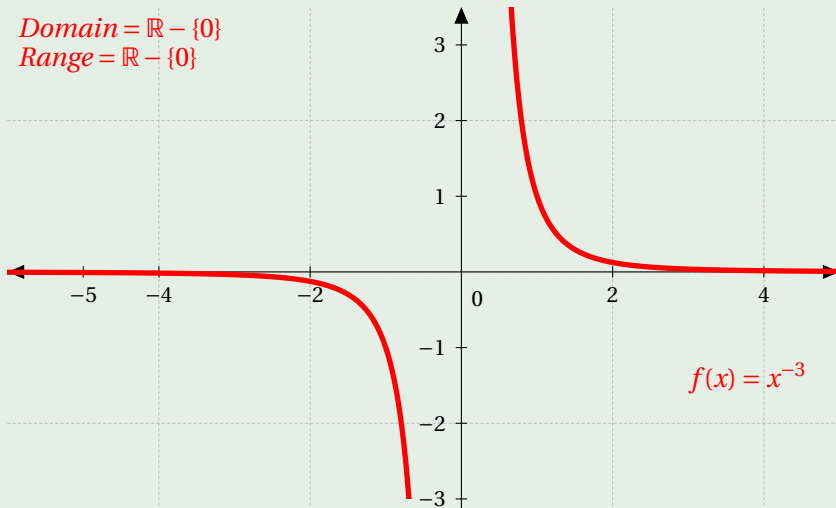
Range = $\mathbb{R}_+ - \{0\}$



Reciprocal functions: Graph of $f(x) = x^{-3}$

Domain = $\mathbb{R} - \{0\}$

Range = $\mathbb{R} - \{0\}$



Radical functions: Graph of $f(x) = \sqrt{x} = x^{1/2}$

$$f(x) = \sqrt{x}$$

$$\text{Domain} = \mathbb{R}_+$$

$$\text{Range} = \mathbb{R}_+$$



Radical functions: Graph of $f(x) = \sqrt[3]{x} = x^{1/3}$

$$g(x) = \sqrt[3]{x}$$

Domain = $\mathbb{R}$
Range = $\mathbb{R}$



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Polynomial Functions

Definition

Sum of monomials of different degrees is called a polynomial. If f is a polynomial, then

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \quad (2)$$

- ① n is a **non-negative integer** called the degree;
- ② a_n is a **non-zero** real number;
- ③ a_i 's are called the coefficients of the polynomial f .



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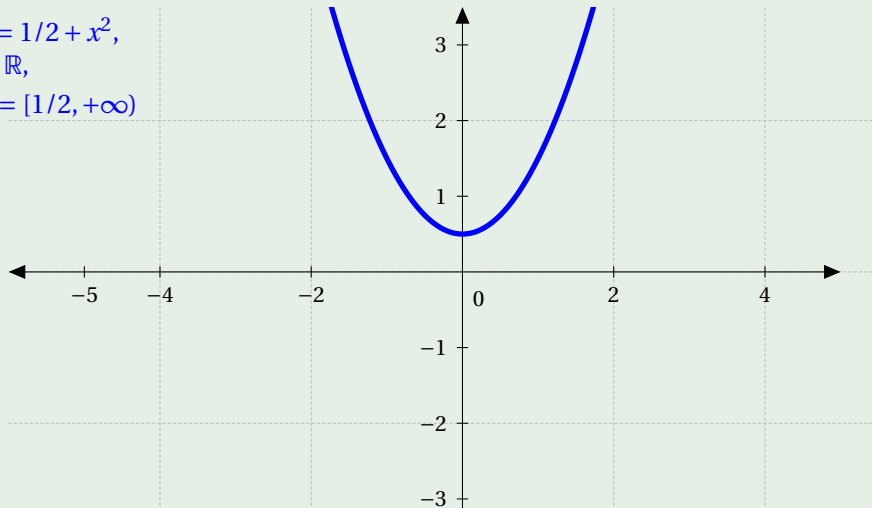
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Note

- ① The domain of a polynomial function is $\mathbb{R}$.
- ② Polynomials of degree 1, 2 and 3 are called **linear**, **quadratic** and **cubic** functions respectively.

Polynomial Functions with 2 as highest power

$$f(x) = 1/2 + x^2,$$
$$D_f = \mathbb{R},$$
$$R(f) = [1/2, +\infty)$$

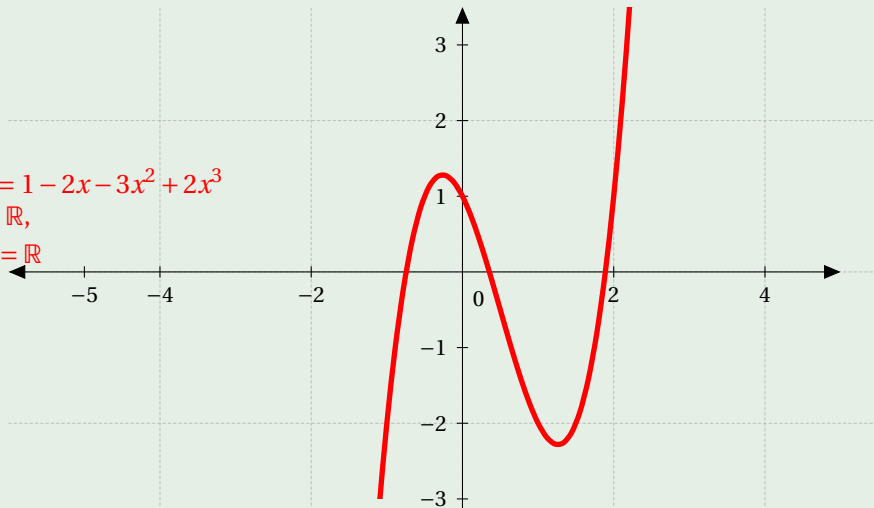


Polynomial Functions with 3 as highest power

$$g(x) = 1 - 2x - 3x^2 + 2x^3$$

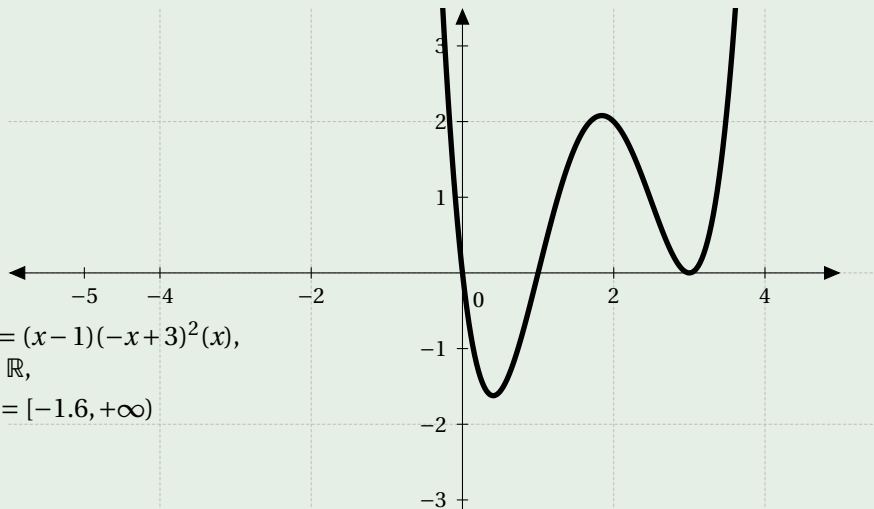
$$D_g = \mathbb{R},$$

$$R(g) = \mathbb{R}$$



Polynomial Functions with 4 as highest power

$$h(x) = (x-1)(-x+3)^2(x),$$
$$D_h = \mathbb{R},$$
$$R(h) = [-1.6, +\infty)$$



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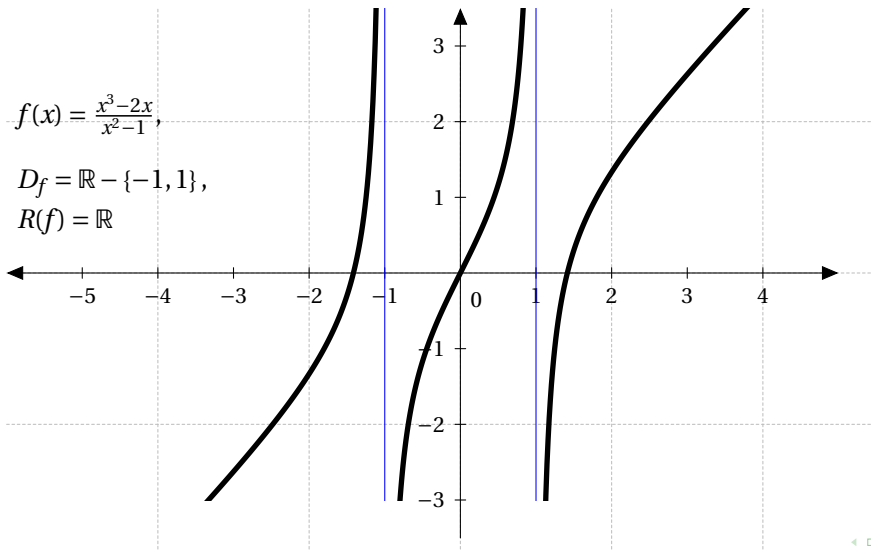
- 1 A **rational function** is a ratio $f(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials.
- 2 The domain of f is $D_f = \{ x \in \mathbb{R} \mid Q(x) \neq 0 \}$.

Example

- 1 $f(x) = \frac{3x}{x^3 - 1}$
- 2 $f(x) = \frac{3x^3 - x^6}{2x^2 - x + 2}$



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- 2 An algebraic function is constructed by taking sums, products, and quotient of polynomials.



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Example

Find the domain of the following algebraic functions:

$$1) f(x) = \sqrt{5 - 2x}$$

$$2) f(x) = \sqrt{x} - \frac{1}{x-1}$$

$$3) f(x) = \frac{2-x}{\sqrt{x-1}-2}$$

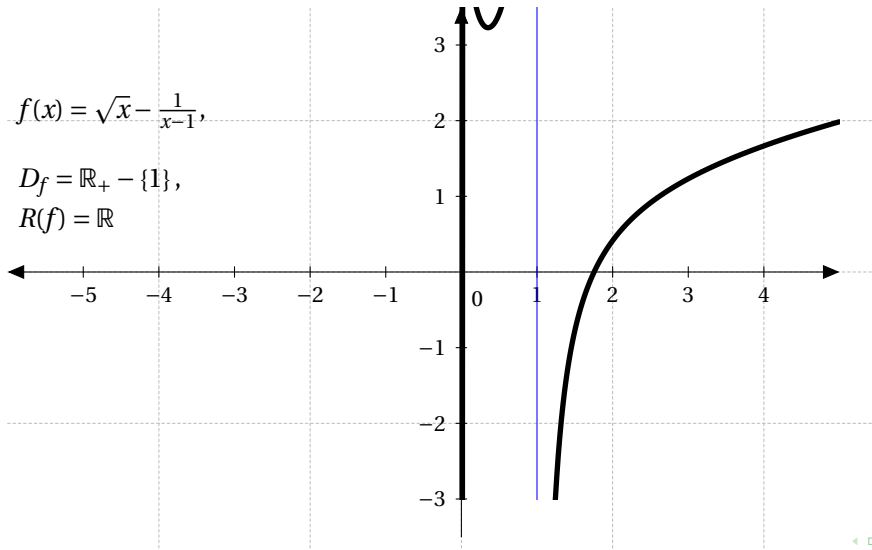


Algebraic functions

$$f(x) = \sqrt{x} - \frac{1}{x-1},$$

$$D_f = \mathbb{R}_+ - \{1\},$$

$$R(f) = \mathbb{R}$$



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- 2 Constant, Step, and Piecewise functions
 - Constant Function
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Transcendental Functions

Definition

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Transcendental functions can be expressed in algebra in terms of an infinite sequence.



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Example

- 1 Exponential functions
- 2 Logarithmic functions
- 3 Trigonometric functions
- 4 Hyperbolic functions
- 5 Inverse of these functions



Exponential Functions

Definition

The function $f(x) = a^x$, where $a > 0$ and $a \neq 1$, is called **exponential function** with base a .

The **domain** of an exponential function is $\mathbb{R}$ and the **range** is $(0, +\infty)$



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- 1) $\left(\frac{2}{3}\right)^x$ 2) 2^x 3) 3^{-x} 4) $\sqrt{7}^x$ 5) e^x 6) e^{-x} .



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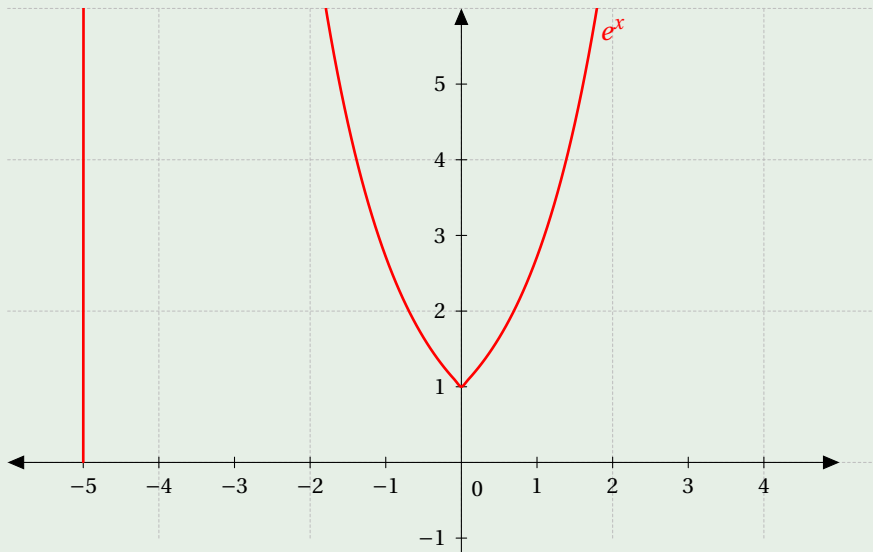
1) $(\frac{2}{3})^x$ 2) 2^x 3) 3^{-x} 4) $\sqrt{7}^x$ 5) e^x 6) e^{-x} .

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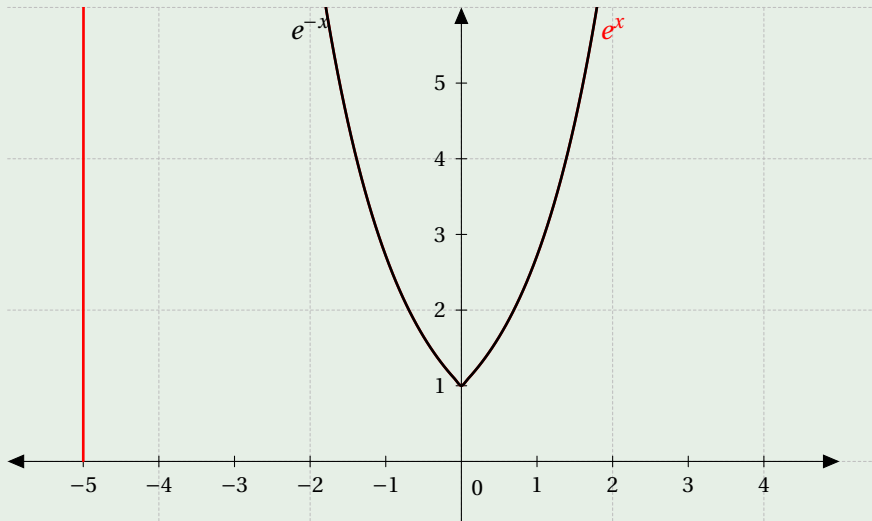
e is mathematical constant called the *Eulernumber* approximated as 2.71828



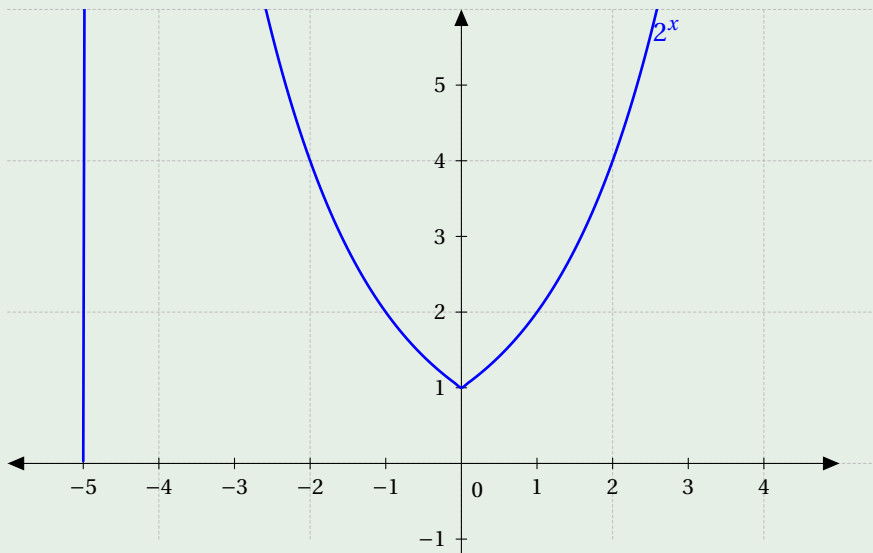
Exponential Functions



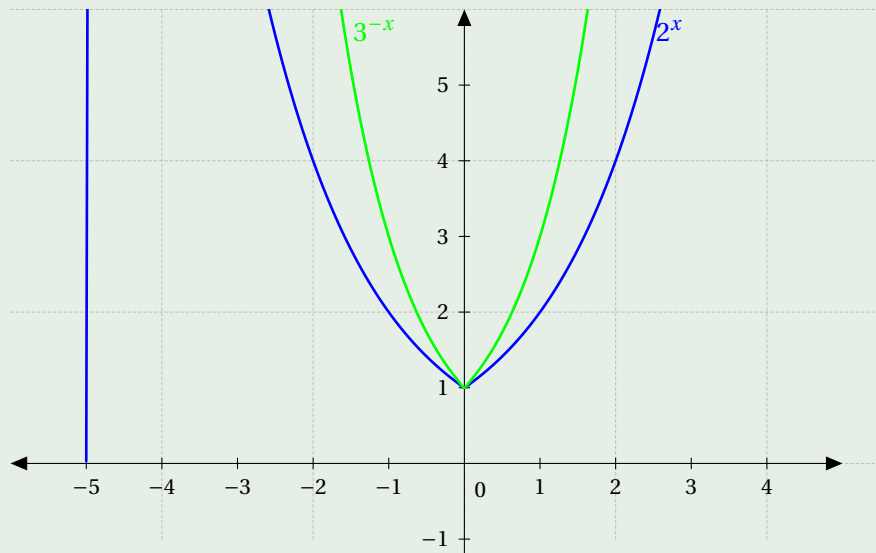
Exponential Functions



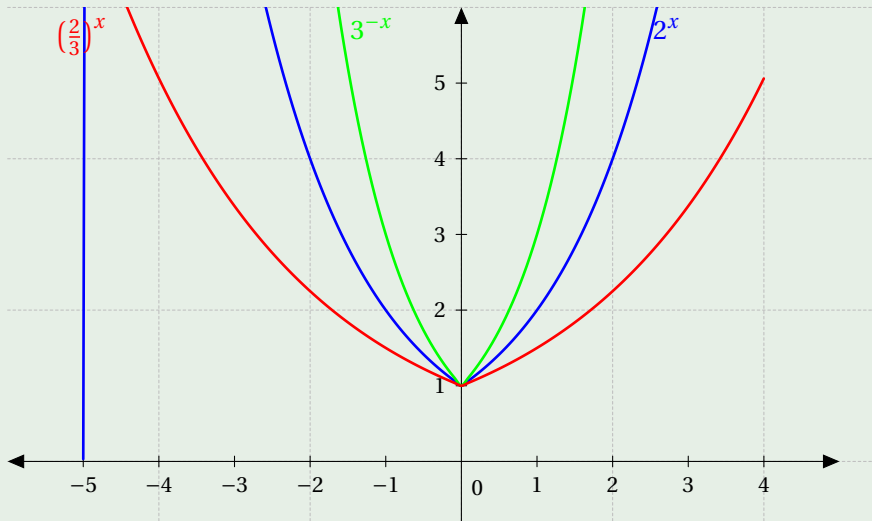
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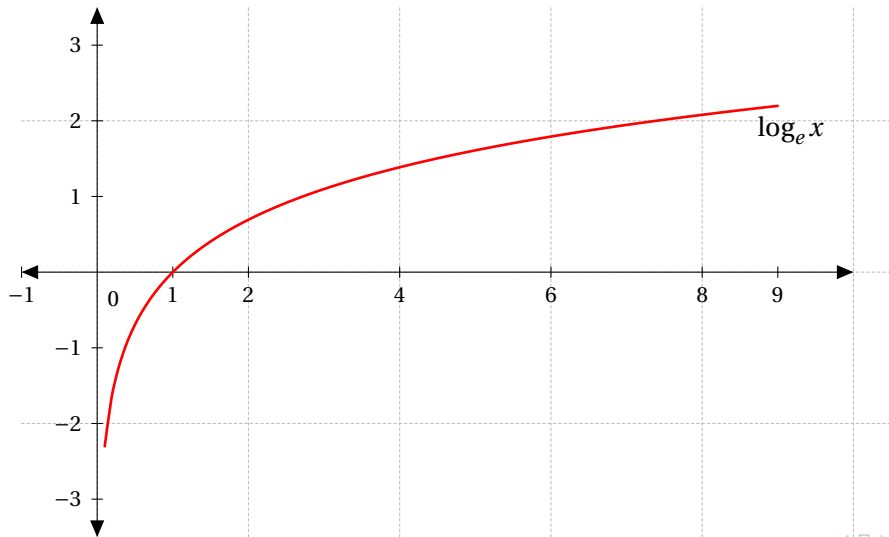
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Natural log

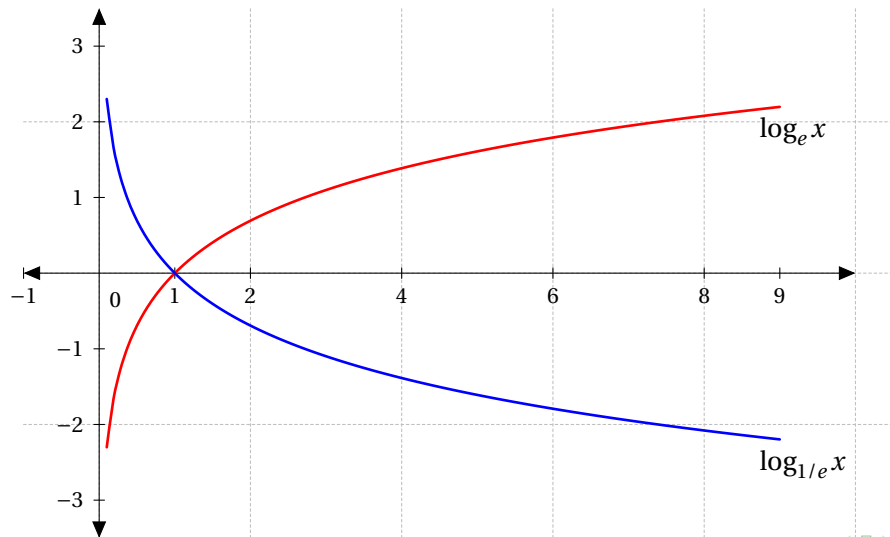
This is the log to the base e and it also called $\ln$. That is

$$\log_e = \ln \quad (3)$$

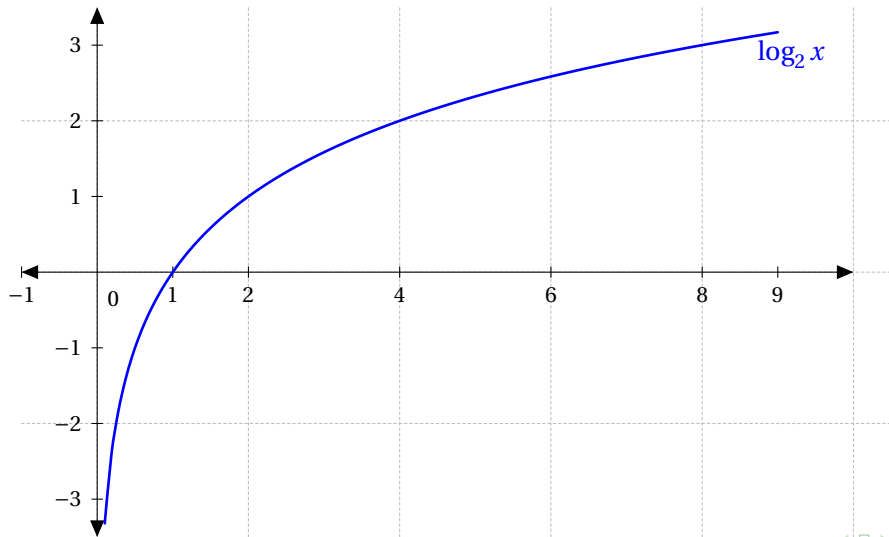
Logarithmic Functions



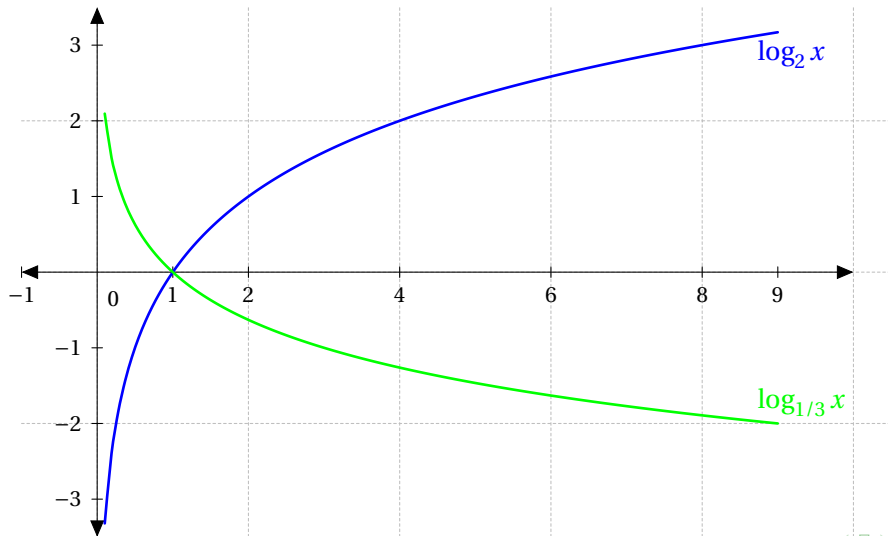
Logarithmic Functions



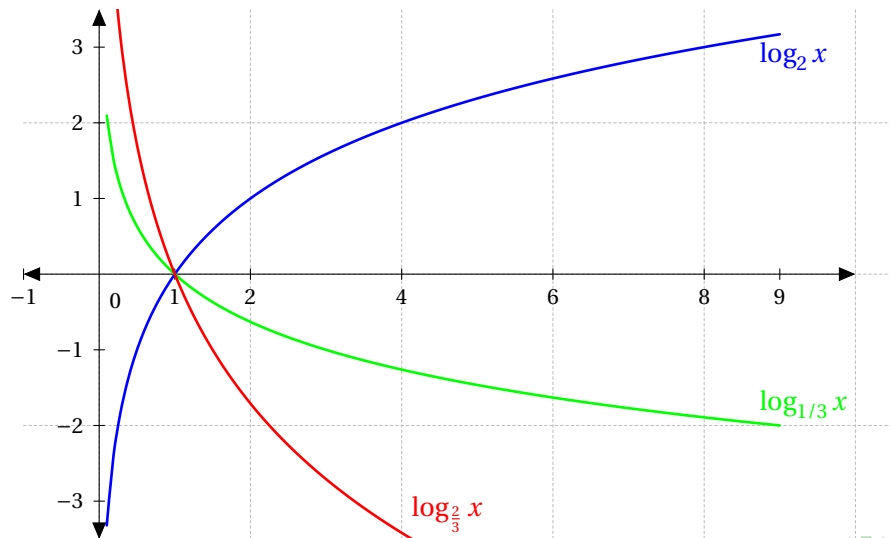
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Some Basic Trig function:

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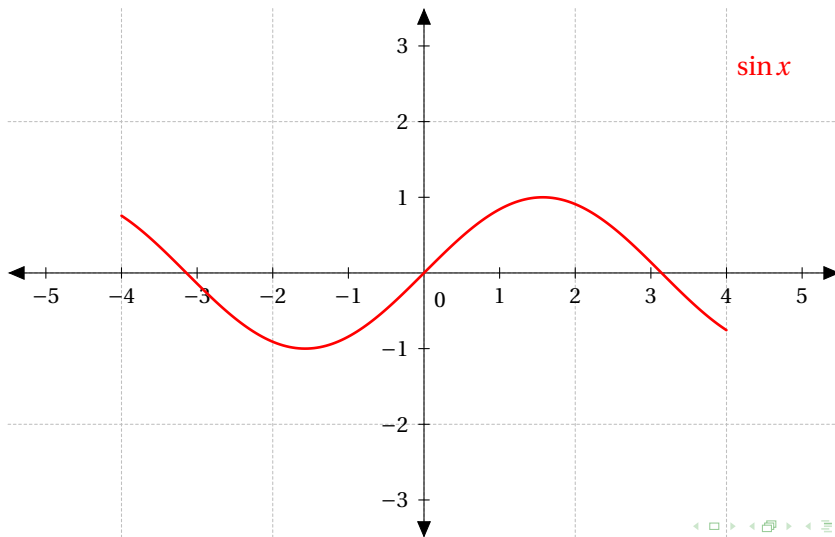
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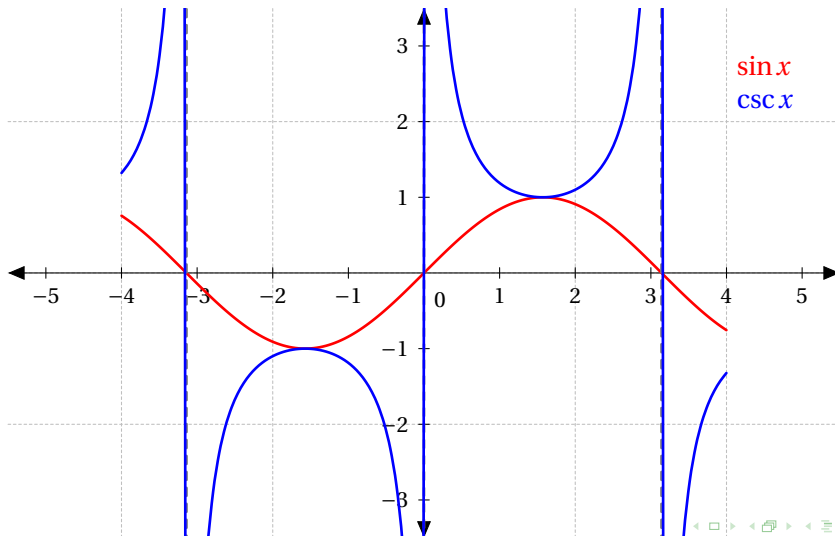
Some Trig Identities

- ① $\cos^2 x + \sin^2 x = 1$
- ② $\sec^2 x - \tan^2 x = 1$
- ③ $\csc^2 x - \cot^2 x = 1$
- ④ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- ⑤ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
- ⑥ $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

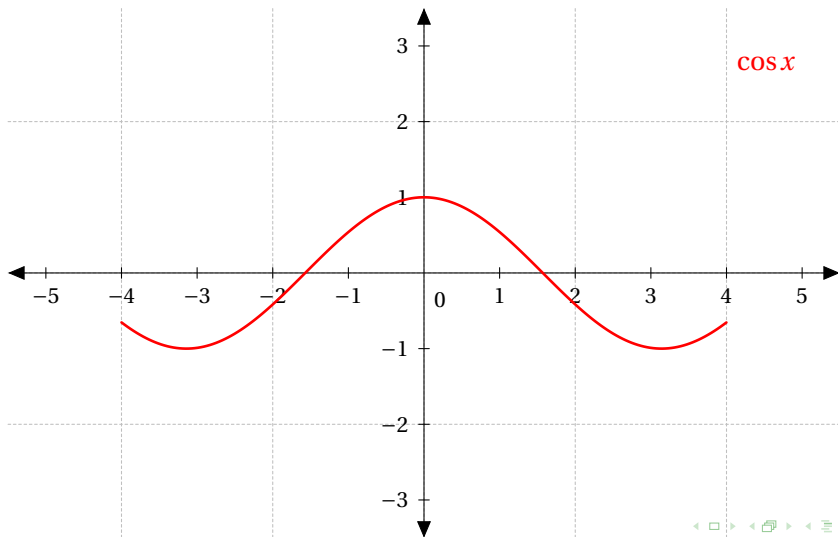
Trigonometric Functions: $\sin$ and $\csc$



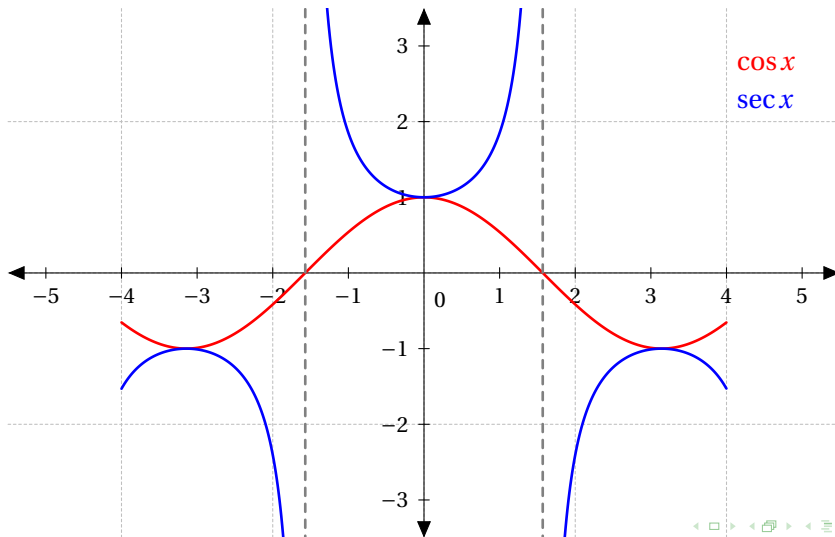
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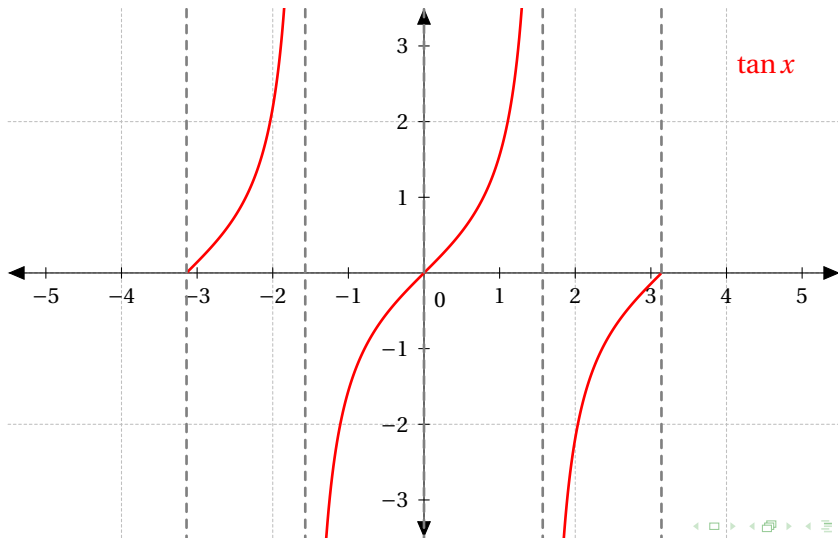
Trigonometric Functions: $\cos$ and $\sec$



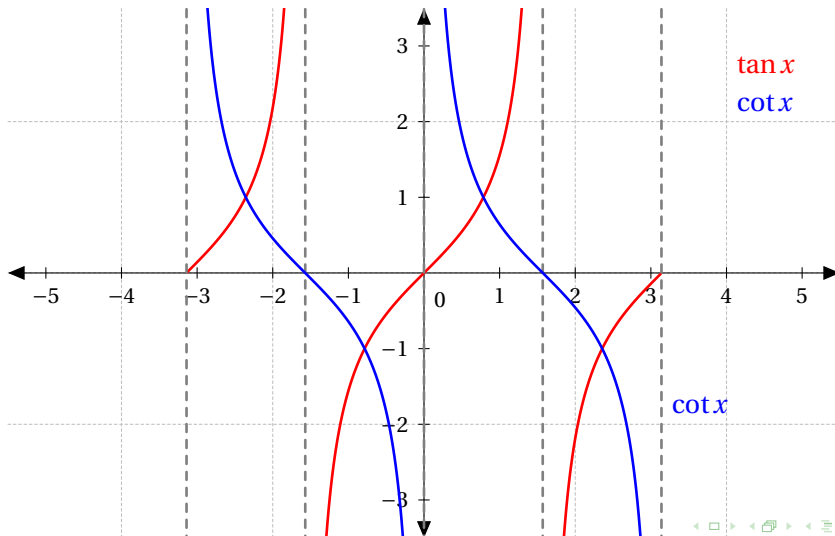
Trigonometric Functions: $\cos$ and $\sec$



Trigonometric Functions: tan and cot



Trigonometric Functions: $\tan$ and $\cot$



Trigonometric Functions

Remarks

$f(x)$	Domain	Range
sin	$\mathbb{R}$	$[-1, 1]$
cos	$\mathbb{R}$	$[-1, 1]$
tan	$\mathbb{R} - \left\{ \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots \right\}$	$\mathbb{R}$
sec	$\mathbb{R} - \left\{ \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots \right\}$	$(-\infty, -1) \cup (1, +\infty)$
csc	$\mathbb{R} - \{ \dots, -2\pi, -\pi, 0, \pi, 2\pi, \dots \}$	$(-\infty, -1) \cup (1, +\infty)$
cot	$\mathbb{R} - \{ \dots, -2\pi, -\pi, 0, \pi, 2\pi, \dots \}$	$\mathbb{R}$

① $D_{\tan} = D_{\sec} = \mathbb{R} - \{ \pi/2 + k\pi \mid k \in \mathbb{Z} \}$

② $D_{\cot} = D_{\csc} = \mathbb{R} - \{ k\pi \mid k \in \mathbb{Z} \}.$



Transcendental: Inverse Trigonometric Functions

There are the functions

- ① $f(x) = \sin^{-1} x$ (also called arc sine or arcsin)
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Note

- ① $y = \sin^{-1} x \Leftrightarrow x = \sin y$
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Transcendental: Hyperbolic and Inverse Hyperbolic Functions

These are functions defined in terms of the exponential functions

Hyperbolic

$$① \sinh x = \frac{e^x - e^{-x}}{2}$$

$$② \cosh x = \frac{e^x + e^{-x}}{2}$$

$$③ \tanh x = \frac{\sinh x}{\cosh x}$$

$$④ \operatorname{csch} x = \frac{1}{\sinh x}$$

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Some identities

$$\textcircled{1} \cosh^2 x - \sinh^2 x = 1$$

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Even Function

Let f be a function and D_f its domain. We assume that if $x \in D_f$ then $-x \in D_f$.

Definition (Even Function)

f is an **even** function if $f(-x) = f(x)$.



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Remarks

The graph of an even function is symmetric about the y-axis.

The graph of an odd function is symmetric about the origin.

Periodic Functions

Definition

Let f be a function, and D_f its domain, then f is a **periodic function** if there exists a **positive** real number t such that $f(x+t) = f(x)$ for all $x \in D_f$.

The **minimum** of such t 's which is often denoted as T , is called the **period** of f .



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Example

The trigonometric functions are periodic functions.

- ① $\sin(x + 2k\pi) = \sin(x + 2\pi) = \sin(x)$ for $k \in \mathbb{Z}$, however, $T = 2\pi$.
- ② $\cos(x + 2k\pi) = \cos(x + 2\pi) = \cos(x)$, $T = 2\pi$.
- ③ $\tan(x + (2k+1)\pi) = \tan(x + \pi) = \tan(x)$, $T = \pi$.



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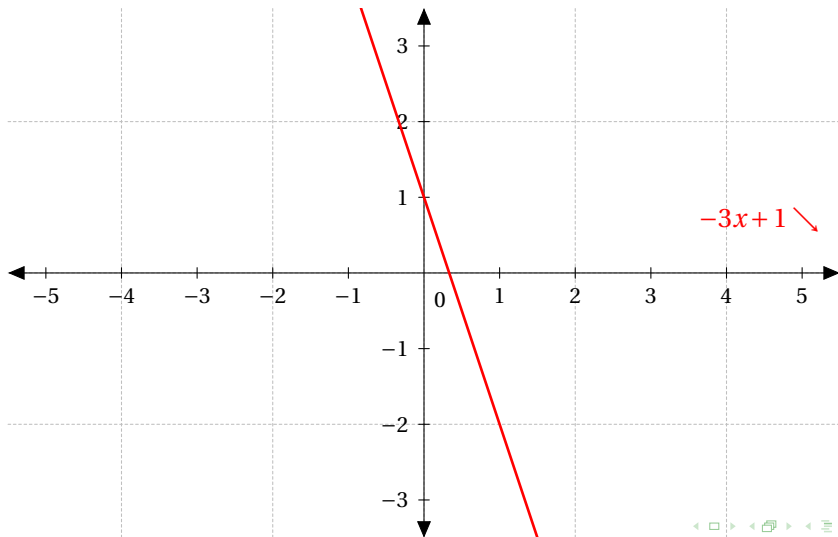
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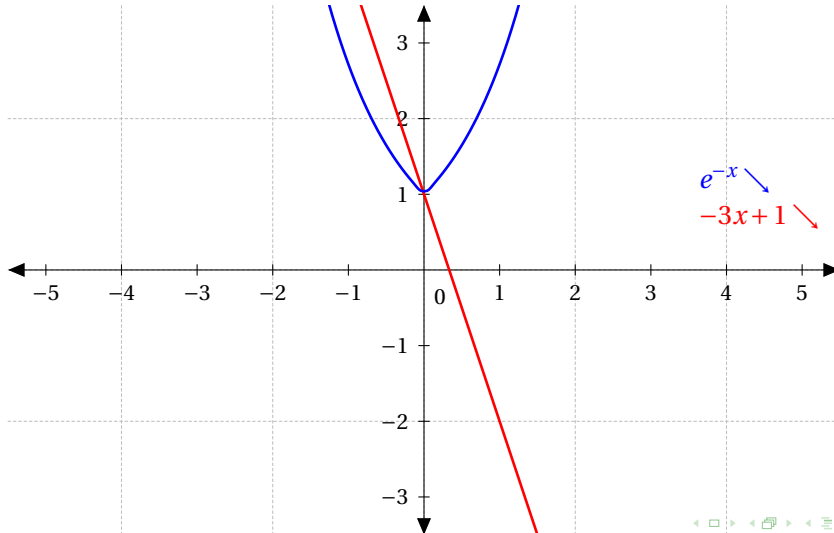
- The functions e^x , $\tan(x)$ and $ax+b$, where $a > 0$, are increasing on their respective domains.
- The functions e^{-x} , $\cot(x)$ and $ax+b$, where $a < 0$, are decreasing on their respective domains.



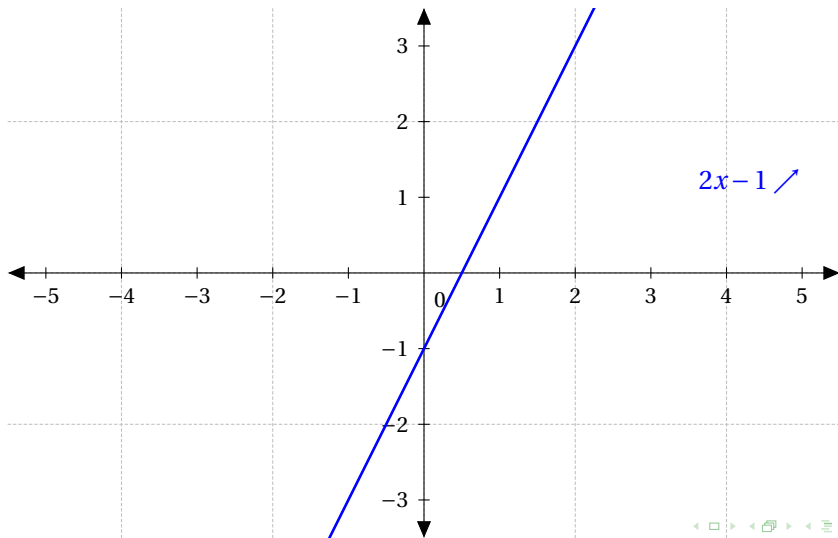
Monotonic Decreasing Function



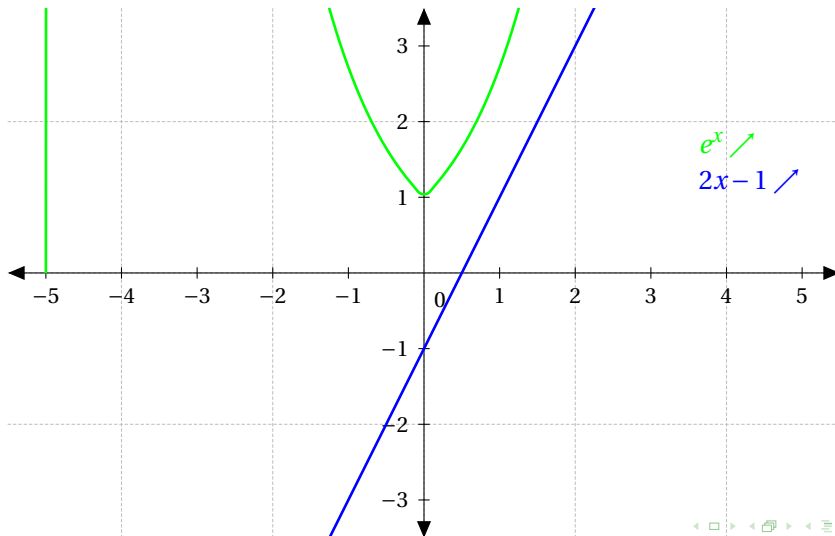
Monotonic Decreasing Function



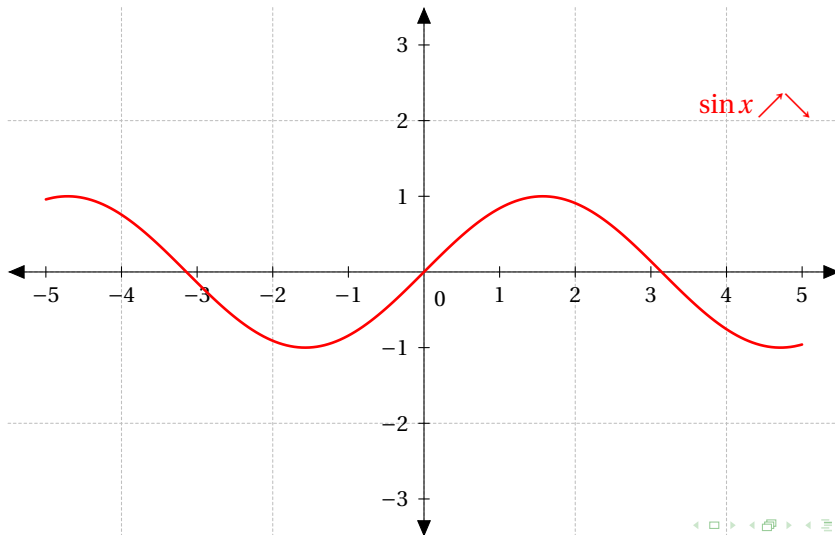
Monotonic Increasing Function



Monotonic Increasing Function



Monotonic Increasing and Decreasing



Monotonic Functions

Example

Show that the function $f(x) = \sqrt{x-2}$ is an increasing function on its domain.



Monotonic Functions

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- ⑤ $\implies f(x_1) < f(x_2)$.
- ⑥ Thus, f is an increasing function on its domain.



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- 2 Constant, Step, and Piecewise functions
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 - Monomials: $p \in \mathbb{Z}_+$
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A function, f , is said to be **bounded below** if there is $\ell \in \mathbb{R}$ such that $f(x) \geq \ell$ for all x in the domain of f .



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Maxima and Minima of Functions

Local(or relative) and Global(or absolute) Minimum

- ① The function f is said to have a **local minimum value** at the point x_0 if $f(x_0) \leq f(x)$ for all x in a **neighbourhood** of x_0 .



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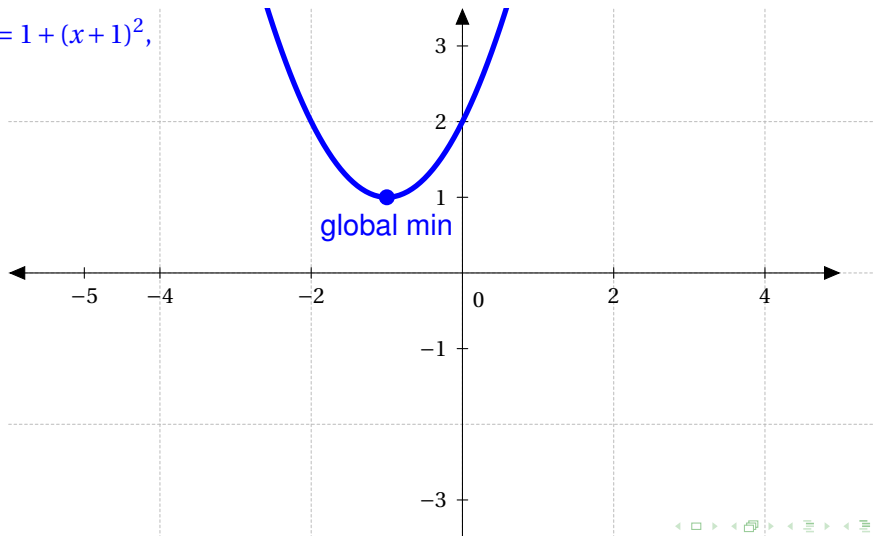
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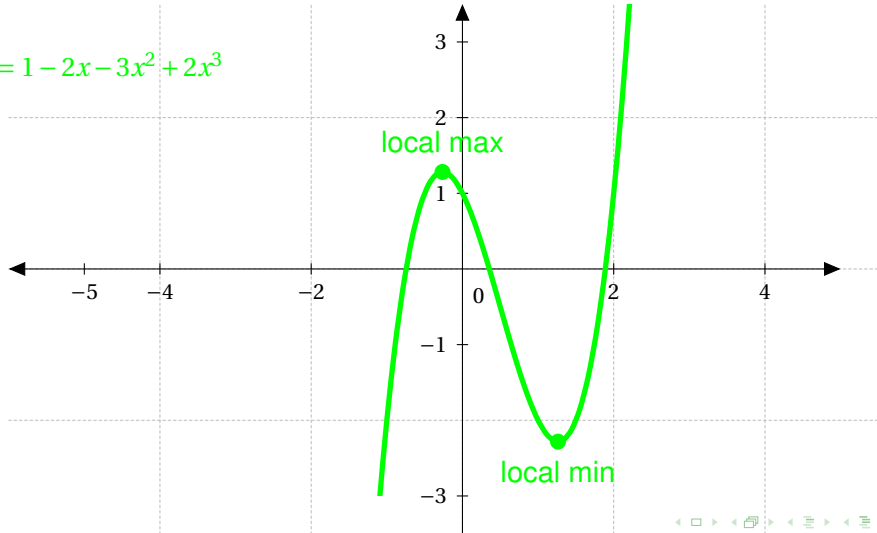
Maxima and Minima of Functions

$$f(x) = 1 + (x + 1)^2,$$

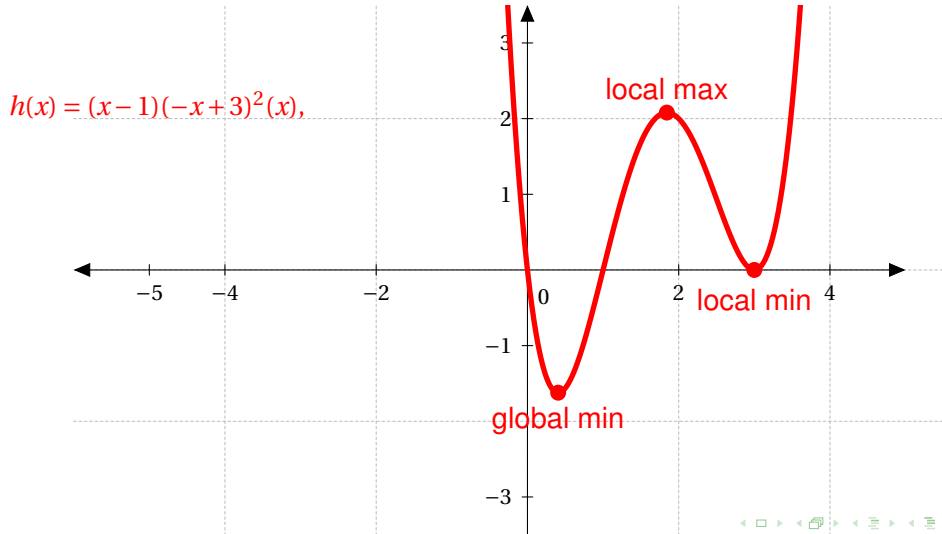


Maxima and minima

$$g(x) = 1 - 2x - 3x^2 + 2x^3$$



Maxima and minima



Exercise

① Which of the following are not polynomial functions?

a. $f(x) = 1$

b. $f(x) = x^2 + x^{-1} + 1$

c. $f(x) = -2x^3 + x^{1/2} - 1$

d. $f(x) = x^4\sqrt{5} - \pi.$

② Find the range of the following polynomial functions:

a. $f(x) = x^2 + 6$

b. $f(x) = -2x^4 - 6$

c. $f(x) = -2x^3 + 1$

d. $f(x) = |-2x^3 + 1|$

e. $f(x) = 3 - 4x, D_f = (-2, 8]$

f. $f(x) = (2x - 1)^2 + 1, D_f = (-\infty, -1) \cup (1, +\infty).$

③ Find the period of the following functions

1) $f(x) = \sin(2x),$ 2) $f(x) = \cos(-2x + \pi/3),$ 3) $f(x) = x - \sin(x).$



Exercise

1 Find the domain of:

$$1) f(x) = \frac{1}{2x-6}$$

$$2) f(x) = \frac{1-x}{1+x}$$

$$3) f(x) = \frac{x^3 - 2x}{x(-x-6)}$$

$$4) f(x) = 3x - 1 - \frac{1}{2x-6}$$

$$5) f(x) = \frac{x}{1-2x+x^2}$$

$$6) f(x) = \frac{x^2 - 2x}{(x-3)(1-x^2)}$$

2 Find the domain of:

$$1) f(x) = \ln x, \quad 2) f(x) = \log_5(1-3x),$$

$$3) f(x) = e^{\frac{1}{x+1}-x}, \quad 4) f(x) = e^{x^2-1} + \ln(|x|+1).$$

3 Determine whether the functions below are even, odd or neither.

$$1) f(x) = e^{x^2-1} + \ln(|x|+1), \quad 2) f(x) = \frac{x^2-2}{x(1-x^2)}, \quad 3) f(x) = x^2 \sin(x)$$

$$4) f(x) = x\sqrt{|x|-1}, \quad 5) f(x) = \ln(\tan x - e^{|x|}), \quad 6) f(x) = x-1.$$



END OF LECTURE
THANK YOU

